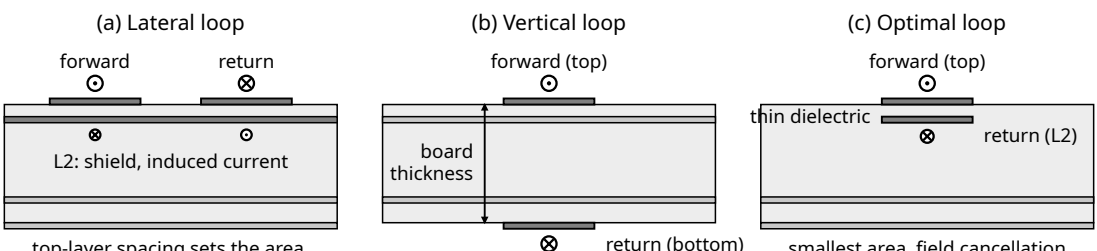
(a) Lateral loop
forward
return
L2: shield, induced current
top-layer spacing sets the area,
eddy currents in the shield reduce it
(b) Vertical loop
forward (top)
board
thickness
return (bottom)
area set by the board thickness
(c) Optimal loop
forward (top)
thin dielectric
return (L2)
smallest area, field cancellation
independent of board thickness